FBXW7-Related Neurodevelopmental Disorder: Comprehensive Disease Characteristics Report

Disease: FBXW7-Related Neurodevelopmental Disorder **MONDO ID:** MONDO:0859280 **Gene:** *FBXW7* (HGNC:13618; NCBI Gene 55294; OMIM gene 606278); chromosome 4q31.3 **Category:** Mendelian, autosomal dominant

Summary

FBXW7-Related Neurodevelopmental Disorder (FBXW7-NDD) is a rare, recently delineated autosomal dominant Mendelian syndrome caused by **monoallelic loss-of-function variants in *FBXW7***, the substrate-recognition subunit of the SCF (SKP1–CUL1–F-box) E3 ubiquitin ligase. The disorder was formally established in 2022 by Stephenson and colleagues, who assembled a cohort of 35 affected individuals harboring de novo and inherited chromosomal deletions, nonsense, frameshift, splice-site, and missense variants (PMID: 35395208). The core clinical phenotype comprises **global developmental delay, borderline-to-severe intellectual disability, hypotonia, and gastrointestinal issues**, with variable structural brain abnormalities affecting the cerebellum, corpus callosum, and white matter. Epilepsy (frequently absence-type), behavioral difficulties, and recurrent respiratory/otolaryngological infections are additional recurring features.

Mechanistically, FBXW7 normally drives phosphorylation-dependent ubiquitination and proteasomal degradation of a set of growth-promoting substrates — most importantly **NOTCH1, CYCLIN E1/E2, c-Jun, c-MYC, mTOR, and BRAF**. Pathogenic variants impair this activity: truncating variants and 4q31.3 deletions reduce protein dosage (haploinsufficiency), WD40-domain missense variants cluster at the substrate-binding surface and reduce substrate affinity, and F-box-domain missense variants disrupt assembly of the SCF complex via impaired SKP1 binding. The downstream consequence — **accumulation of undegraded substrates and aberrant NOTCH/c-Jun/MYC signaling** — dysregulates neural stem cell proliferation, differentiation, and migration. This mechanism is strongly corroborated across evolutionarily conserved model systems (mouse, zebrafish, *Drosophila*).

Clinically important secondary features include a **Wilms tumor predisposition** operating through a classic two-hit tumor-suppressor mechanism (germline first hit plus a somatic second hit in renal tissue), which warrants renal ultrasound surveillance in individuals with constitutional/germline variants. The disorder shows **incomplete penetrance and variable, partly domain-dependent expressivity** — F-box-domain variants can produce a strikingly mild phenotype (isolated drug-responsive absence epilepsy without intellectual disability), whereas WD40-domain variants and structural lesions tend to produce more severe, multisystem presentations. Management is currently supportive: early developmental therapies, anti-seizure medication, management of constipation and infections, tumor surveillance, and genetic counseling. No targeted or disease-modifying therapy exists.

Section 1 — Disease Information

Overview. FBXW7-NDD is a monogenic neurodevelopmental disorder caused by heterozygous loss-of-function of *FBXW7*, which encodes the F-box/WD-repeat protein that serves as the substrate-recognition component of an SCF-type Cullin-RING E3 ubiquitin ligase. Because *FBXW7* is a canonical tumor suppressor, the disorder bridges two domains: a neurodevelopmental syndrome and a cancer (Wilms tumor) predisposition.

Key identifiers. - **MONDO:** MONDO:0859280 - **Gene:** *FBXW7* — HGNC:13618; NCBI Gene 55294; OMIM gene 606278; locus 4q31.3 - **OMIM / ICD / Orphanet / MeSH:** Specific OMIM phenotype, ICD-10/ICD-11, and Orphanet numbers were not definitively confirmed within this investigation and should be verified against OMIM and Orphanet directly. The disorder is sometimes catalogued under broad "autosomal dominant intellectual disability" ICD rubrics.

Synonyms / alternative names. FBXW7 neurodevelopmental syndrome; FBXW7-related neurodevelopmental disorder; FBX(W7)-related neurodevelopmental syndrome (as used in the CNV literature). The gene itself has aliases including *CDC4*, *SEL10*, *hCDC4*, and *FBW7*.

Information source. The knowledge base is derived from **aggregated disease-level resources** — principally a multicenter cohort study (PMID: 35395208), subsequent multicenter case series (PMID: 42432493), a dedicated review (PMID: 42111496), individual case reports, and model-organism studies — rather than from individual EHR mining.

Section 2 — Etiology

Primary cause — genetic. FBXW7-NDD is caused by **monoallelic (heterozygous) loss-of-function variants in *FBXW7***. Stephenson et al. documented the full loss-of-function spectrum: chromosomal deletions, nonsense, frameshift, splice-site, and missense variants, arising both de novo and inherited (PMID: 35395208): *"35 individuals harboring de novo and inherited FBXW7 germline monoallelic chromosomal deletions and nonsense, frameshift, splice-site, and missense variants associated with a neurodevelopmental syndrome."* Structural/regulatory lesions at 4q31.3 (deletions, duplications, translocations with position effects) are an additional causal class (see Section 4).

Genetic risk factors. The causal variant itself is the risk factor; there are no established susceptibility loci or polygenic contributors beyond the primary *FBXW7* lesion. Domain location of missense variants modulates severity (modifier-like effect — see Section 9). A somatic *FBXW7* second-hit variant is the proximate risk factor for Wilms tumor in germline carriers.

Environmental risk factors. None established. The disorder is Mendelian; no toxin, infectious, lifestyle, or occupational exposure has been linked to causation. Many cases are de novo, implicating spontaneous germline mutation (for which advanced parental age is a general, non-specific risk factor for de novo mutation).

Protective factors. None identified (genetic or environmental). This has not been systematically studied given the rarity of the disorder.

Gene–environment interactions. None documented. The two-hit tumor mechanism is a gene–(somatic)gene interaction rather than a gene–environment interaction.

Section 3 — Phenotypes

FBXW7-NDD is a multisystem but neurologically-dominant disorder. The table below consolidates phenotype frequencies, drawing principally on the 2026 multicenter case series of 7 individuals (PMID: 42432493) and the foundational cohort (PMID: 35395208).

Phenotype	Type	Suggested HPO term	Frequency (case series)	Onset / severity
Hypotonia	Clinical sign	HP:0001252	7/7 (100%)	Neonatal/ infantile; variable
Global developmental delay	Clinical sign	HP:0001263	6/7 (~86%)	Early childhood
Language/speech impairment	Clinical sign	HP:0000750	6/7 (~86%)	Childhood
Intellectual disability	Clinical sign	HP:0001249	4/7 (~57%); borderline–severe	Childhood; variable
Epileptic seizures (incl. absence)	Clinical sign	HP:0001250 / HP:0002121	4/7 (~57%)	Childhood; often drug-responsive
Corpus callosum anomaly	Imaging/ structural	HP:0001273	2/6 on MRI	Congenital
Cerebellar abnormality	Imaging/ structural	HP:0001317	Variable	Congenital
White matter abnormality	Imaging/ structural	HP:0002500	Variable	Congenital
Constipation	Symptom (GI)	HP:0002019	Common extracerebral feature	Variable
Recurrent respiratory/ENT infections	Clinical sign	HP:0002788	Common extracerebral feature	Childhood
Behavioral difficulties	Behavioral	HP:0000708	Variable (can predominate)	Childhood
Wilms tumor (predisposition)	Neoplasm	HP:0002667	Rare but recurrent	Childhood

From the case series (PMID: 42432493): *"Neurodevelopmental involvement was present in six individuals and was characterized by global developmental delay and language impairment; hypotonia was observed in all seven. Formal intellectual disability was documented in four cases."* Epilepsy frequency: *"Epileptic seizures occurred in four individuals, whereas three had no history of epilepsy."* Extracerebral features: *"Extracerebral features were variably reported, most commonly constipation and recurrent respiratory/otolaryngological infections."*

Severity and progression. The neurodevelopmental deficits are **static/non-progressive** (a developmental, not neurodegenerative, disorder), though functional impact is lifelong. Severity is **highly variable**, ranging from preserved cognition with isolated behavioral difficulties or isolated absence epilepsy, to severe intellectual disability with structural brain anomalies. The 2022 cohort described the spectrum as *"borderline to severe intellectual disability"* (PMID: 35395208).

Quality-of-life impact. No disease-specific EQ-5D/SF-36/PROMIS data exist. Impact is inferred from phenotype: intellectual disability, language impairment, and epilepsy substantially affect daily functioning, education, communication, and independence; hypotonia and GI issues add care burden. Milder (F-box-variant) individuals may have near-normal functioning.

Section 4 — Genetic / Molecular Information

Causal gene. *FBXW7* (F-box and WD repeat domain containing 7); HGNC:13618; NCBI Gene 55294; OMIM gene 606278; locus **4q31.3**. The protein has three functional regions relevant to disease: an N-terminal dimerization domain, an **F-box domain** (binds SKP1 to assemble the SCF complex), and a C-terminal **WD40 β-propeller domain** (binds phosphorylated substrates via a conserved phosphodegron, CPD).

Variant spectrum and classification. All variant classes consistent with loss of function have been reported — **nonsense, frameshift, splice-site, missense, and whole/partial-gene deletions** — most classified pathogenic/likely pathogenic per ACMG/AMP criteria (PMID: 35395208). Representative missense: c.926G>C p.(Arg309Pro) in the F-box domain (PMID: 41685408).

Functional consequences — loss of function (haploinsufficiency). Several lines of evidence converge on loss of function: - Truncating variants and deletions reduce gene dosage. - WD40 missense variants cluster at the substrate-binding surface: *"A crystal-structure model of FBXW7 predicted that missense variants were clustered at the substrate-binding surface of the WD40*

domain and that these might reduce FBXW7 substrate binding affinity" (PMID: 35395208). - Direct functional assay: *"Expression of recombinant FBXW7 missense variants in cultured cells demonstrated impaired CYCLIN E1 and CYCLIN E2 turnover"* (PMID: 35395208). - Position-effect lesion: *"FBXW7 expression in the immortalized lymphoblasts of the patient was reduced compared to that in controls"* (PMID: 39123068).

Allele frequency. Pathogenic variants are absent or ultra-rare in gnomAD, consistent with a highly penetrant, deleterious disorder; *FBXW7* is strongly constrained against loss-of-function variation (high pLI), supporting haploinsufficiency as the disease mechanism.

Somatic vs germline. FBXW7-NDD is caused by **germline (constitutional)** variants. In Wilms tumor, a **somatic second-hit** variant on the other allele is detected in tumor DNA, consistent with Knudson's two-hit model (PMID: 39414990). *FBXW7* is also one of the most commonly somatically mutated tumor-suppressor genes across sporadic cancers.

Modifier genes. No formal modifier genes identified. The principal determinant of expressivity appears to be the **affected protein domain** (F-box vs WD40 — see Section 9).

Epigenetic information. No disease-specific DNA-methylation episignature has been established for FBXW7-NDD. (In cancer contexts FBXW7 can be downregulated epigenetically, but this is not documented as a germline-disorder mechanism.)

Chromosomal abnormalities. Large-scale lesions at 4q31.3 cause the disorder: - De novo **deletion and duplication** affecting the FBXW7 regulatory region/TAD boundary, with Hi-C-confirmed rewiring in neural progenitors (PMID: 38780184): *"two patients with a de novo deletion and a duplication in chromosome 4q31, potentially causing FBX-related neurodevelopmental syndrome by affecting the regulatory region of FBXW7."* - A de novo **translocation t(3;4)(q27;q31.2)** producing a downstream position effect that reduced FBXW7 expression (PMID: 39123068).

Section 5 — Environmental Information

Environmental factors. None established as causal or contributory. FBXW7-NDD is a Mendelian disorder.

Lifestyle factors. Not applicable to disease causation.

Infectious agents. None. (Recurrent respiratory/ENT infections are a *consequence/feature* of the disorder, not a cause.)

Section 6 — Mechanism / Pathophysiology

Ordered causal chain

1. A **monoallelic loss-of-function lesion in *FBXW7*** (truncating variant, WD40 missense, F-box missense, 4q31.3 deletion/duplication, or position-effect translocation) **reduces functional FBXW7 protein or impairs its activity**.
2. *Branch A (truncating / deletion / position effect):* reduced FBXW7 **dosage** (haploinsufficiency).
3. *Branch B (WD40 missense):* intact protein with **reduced substrate-binding affinity** (crystal-structure modeling; PMID: 35395208).
4. *Branch C (F-box missense):* impaired **SKP1 binding → defective SCF complex assembly** (PMID: 41685408).
5. Impaired FBXW7 function **results in decreased phosphodegron-dependent ubiquitination** of its substrates by the SCF(FBXW7) E3 ligase.
6. This **leads to reduced proteasomal degradation and accumulation** of substrate oncoproteins — **NOTCH1, CYCLIN E1/E2, c-Jun, c-MYC, mTOR, BRAF, MCL-1** (demonstrated for CYCLIN E1/E2 in patient variants; demonstrated for NOTCH1 and c-Jun in mouse).
7. Substrate accumulation **results in aberrant activation of downstream signaling** — chiefly **NOTCH signaling** and **c-Jun/MYC/mTOR-driven proliferation** (with cross-talk to PI3K/AKT and Wnt/β-catenin).
8. Dysregulated signaling **disrupts neural stem/progenitor cell proliferation, differentiation, migration, and apoptosis** (demonstrated in mouse cerebellum and zebrafish spinal cord; *inferred* for human brain).
9. This **leads to the developmental structural and functional CNS defects** — cerebellar hypoplasia/Purkinje cell loss, corpus callosum and white-matter anomalies, abnormal cortical/neuronal function.
10. These **manifest clinically** as global developmental delay, intellectual disability, hypotonia, epilepsy, and behavioral difficulties.
11. *Parallel branch (tumor):* in renal (and other) tissue, the germline first hit plus a **somatic second-hit *FBXW7* variant** causes biallelic inactivation → unchecked CYCLIN E/NOTCH-driven proliferation → **Wilms tumor**.

Supporting detail

Molecular pathways. FBXW7 is *"a member of the F-box protein family within the ubiquitin-proteasome system [that] performs an indispensable role in orchestrating cellular processes through ubiquitination and degradation of its substrates, such as c-MYC, mTOR, MCL-1, Notch, and cyclin E"* (PMID: 35515121). It *"serves a role in phosphorylation-dependent ubiquitination and proteasome degradation of oncoproteins"* (PMID: 35346215). Dysregulated pathways therefore include **NOTCH, MYC, mTOR, Wnt/β-catenin, and PI3K/AKT**. Suggested GO terms: GO:0016567 (protein ubiquitination), GO:0043161 (proteasome-mediated ubiquitin-dependent protein catabolic process), GO:0007219 (Notch signaling pathway), GO:0008285 (negative regulation of cell proliferation).

Cellular processes. Cell-cycle regulation (via CYCLIN E), neural stem cell differentiation vs self-renewal (via NOTCH), apoptosis and stress response (via c-Jun, MCL-1), and progenitor migration. *"Fbw7 is highly expressed in the nervous system and controls neural stem cell differentiation and apoptosis via Notch and c-Jun during embryonic development"* (PMID: 21827743).

Protein dysfunction. Loss of function dominates: reduced abundance (truncating/deletion), reduced substrate binding (WD40 missense), or failed SCF assembly (F-box missense). FBXW7 functions as a dimer; some missense variants may additionally act in a dominant-negative fashion on the wild-type allele, although the predominant model is haploinsufficiency/loss of function.

Subcellular localization. FBXW7 isoforms localize to the **nucleus (α), cytoplasm/nucleoplasm (β), and nucleolus (γ)**; substrate degradation occurs via the **26S proteasome**. Suggested GO cellular component terms: GO:0019005 (SCF ubiquitin ligase complex), GO:0005634 (nucleus), GO:0005730 (nucleolus).

Metabolic / immune / biochemical. No specific metabolic defect or primary immune abnormality is established. Recurrent infections are likely secondary (e.g., hypotonia-related aspiration, structural ENT factors) rather than a primary immunodeficiency. The core biochemical defect is deficient E3-ligase-mediated substrate turnover.

Model-organism evidence (cross-species convergence on NOTCH). - **Mouse:** Conditional cerebellar *Fbxw7* knockout → *"decreased cerebellar size, reduced Purkinje cell number and defects in axonal arborisation … supranumeral fissures and aberrant progenitor cell migration"*, with *"Notch1 and N-terminally phosphorylated c-Jun … upregulated"* (PMID: 21827743). Concomitant c-Jun deletion rescued Purkinje cell number. - **Zebrafish:** *"Notch signaling activity was elevated in spinal cord precursors of fbxw7 mutant zebrafish and inhibition of Notch*

signaling suppressed formation of excess OPCs" (PMID: 22554084). - **Drosophila:** *"Pan-neuronal knockdown of the Drosophila ortholog, archipelago, impaired learning and neuronal function"* (PMID: 35395208).

Cell types involved (CL suggestions): neural stem/progenitor cells (CL:0000047), radial glial cells (CL:0000681), cerebellar Purkinje cells (CL:0000121), oligodendrocyte precursor cells (CL:0002453), neurons (CL:0000540).

Section 7 — Anatomical Structures Affected

Organ / system level. The **central nervous system (UBERON:0001017)** is the primary affected system. Within the brain, affected structures include: - **Cerebellum (UBERON:0002037)** — hypoplasia, supernumerary fissures (strong mouse and human evidence). - **Corpus callosum (UBERON:0002336)** — thinning/anomalies (2/6 on MRI in the case series). - **Cerebral white matter (UBERON:0002316)** — abnormalities. - **Cerebral cortex (UBERON:0000956)** — functional involvement (epilepsy, intellectual disability).

Secondary organ / system involvement. Digestive system (UBERON:0001007) — constipation; **respiratory/ENT (UBERON:0001004)** — recurrent infections; **kidney (UBERON:0002113)** — Wilms tumor (nephroblastoma arising from renal blastema). The 2024 CNV cases additionally reported **cardiac and limb** abnormalities and facial dysmorphism (PMID: 38780184), and a translocation case showed growth failure and skeletal manifestations (PMID: 39123068).

Tissue / cell level. Nervous tissue is primarily affected — neural stem/progenitor cells, radial glia, cerebellar Purkinje cells, and oligodendrocyte precursor cells (see CL terms above). In Wilms tumor, renal blastemal (metanephric) cells are the neoplastic population.

Subcellular level. Nucleus, nucleolus, cytoplasm, and the SCF ubiquitin-ligase/proteasome machinery (GO:0019005, GO:0005634, GO:0005730).

Lateralization. CNS involvement is **bilateral** (developmental, midline, and generalized), not lateralized.

Section 8 — Temporal Development

Onset. Congenital / early childhood. Structural brain anomalies are present from birth; hypotonia is typically neonatal/infantile; developmental delay becomes evident in infancy/early childhood; epilepsy onset is typically in childhood (e.g., early-onset childhood absence epilepsy in the F-box-variant case; PMID: 41685408). Onset pattern is **insidious/developmental**, not acute.

Progression. The disorder is **static (non-progressive)** at the neurological level — a neurodevelopmental, not neurodegenerative, condition. Deficits persist lifelong but do not typically worsen mechanistically; developmental trajectories reflect delay rather than regression. Epilepsy may be episodic and is often drug-responsive. Wilms tumor, if it occurs, follows the childhood timeline typical of nephroblastoma.

Patterns / critical periods. The critical period of vulnerability is **embryonic/early postnatal neurogenesis**, when FBXW7-dependent control of neural progenitor proliferation and differentiation is essential (mouse neural-lineage deletion is perinatal lethal; PMID: 21827743). This also defines the window in which the structural brain phenotype is established, before clinical diagnosis is typically possible.

Section 9 — Inheritance and Population

Epidemiology. FBXW7-NDD is **ultra-rare**; precise prevalence/incidence are not established. Fewer than ~50 individuals are reported in the literature across the foundational cohort (35; PMID: 35395208), the 7-individual case series (PMID: 42432493), and additional single-case reports. No Orphanet point-prevalence figure was confirmed in this investigation.

Inheritance. Autosomal dominant, monoallelic. Most cases are **de novo** (e.g., 6/7 de novo in the case series; PMID: 42432493), but **inherited** cases from mildly affected or apparently unaffected parents occur (PMID: 35395208).

Penetrance and expressivity. Transmission from mildly/unaffected parents and a reported constitutional mosaic variant indicate **incomplete penetrance and variable expressivity**. Expressivity is partly **domain-dependent**:

Variant class / domain	Representative variant	Phenotype severity	Molecular mechanism
WD40-domain missense	clustered substrate-binding surface variants	More severe (ID, structural brain anomalies)	Reduced substrate-binding affinity (PMID: 35395208)
F-box-domain missense	c.926G>C p.(Arg309Pro)	Milder (isolated drug-responsive absence epilepsy, no ID)	Impaired SKP1 binding/SCF assembly (PMID: 41685408)
Truncating / deletion	nonsense, frameshift, 4q31.3 del	Variable; often significant	Haploinsufficiency (PMID: 35395208)
Regulatory / structural	4q31.3 CNV, t(3;4) position effect	Severe, multisystem (cardiac, limb, skeletal, growth)	Reduced/dysregulated FBXW7 expression (PMID: 38780184, PMID: 39123068)

The F-box case author's summary captures the contrast: *"unlike prior cases predominantly associated with WD40 domain variants and severe phenotypes, our patient exhibited a much milder clinical presentation consisting of isolated, drug-responsive absence seizures without intellectual disability"* (PMID: 41685408).

Genetic anticipation. Not applicable (not a repeat-expansion disorder). **Germline mosaicism:** possible (a constitutional mosaic variant has been reported); relevant to recurrence-risk counseling. **Founder effects / consanguinity / carrier frequency:** not applicable — a dominant disorder, predominantly de novo, with no reported founder variants.

Population demographics. No ethnic/geographic predilection established; cases are reported worldwide (Europe, North America, Asia, India). **Sex ratio:** no strong skew documented (case series: 5 male, 2 female — too small to infer). **Age distribution:** primarily identified in pediatric populations.

Section 10 — Diagnostics

Diagnostic approach — molecular. Diagnosis is genetic. **Trio exome sequencing (WES) or genome sequencing (WGS)** is first-line and detects SNVs/indels; because many cases are de novo, trio testing maximizes yield. **Chromosomal microarray (CMA)** detects 4q31.3 deletions/

duplications (PMID: 38780184). **Karyotyping / long-read WGS** is needed for cryptic balanced rearrangements and position effects (PMID: 39123068). Single-gene *FBXW7* testing is appropriate when the phenotype is strongly suggestive. The gene appears on broad intellectual-disability gene panels used in neurodevelopmental cohorts (PMID: 38114583).

Clinical/laboratory tests and biomarkers. There is **no specific biochemical biomarker** and **no newborn-screening test**. Research assays (e.g., CYCLIN E1/E2 turnover in patient cells, lymphoblast *FBXW7* expression) support variant interpretation but are not routine. Immunohistochemistry can show mosaic/loss of FBXW7 protein in tumor tissue.

Imaging / electrophysiology. Brain MRI characterizes cerebellar, corpus callosum, and white-matter anomalies (though it is normal in a substantial minority — 4/6 in the case series; PMID: 42432493). **EEG** characterizes seizures (e.g., generalized spike-wave in absence epilepsy). **Renal/abdominal ultrasound** is used for tumor surveillance.

Clinical criteria / differential diagnosis. There are no formal consensus diagnostic criteria beyond molecular confirmation in a compatible phenotype. Differential diagnosis includes other monogenic intellectual-disability/hypotonia syndromes and, given the NOTCH/MYC axis, syndromes affecting the same pathway (e.g., *MYCN*-related megalencephaly-polydactyly, which acts through FBXW7-dependent MYCN degradation; PMID: 37710961).

Screening. Cascade testing of at-risk relatives after a proband variant is identified; tumor surveillance (below) functions as secondary prevention.

Section 11 — Outcome / Prognosis

Survival / mortality. No disease-specific mortality or survival data are available. The neurodevelopmental disorder itself is **not inherently life-limiting**; life expectancy is most influenced by severity of complications (e.g., epilepsy control, aspiration/infection risk) and by Wilms tumor outcomes (nephroblastoma generally has favorable survival with modern therapy, though this depends on stage and histology).

Morbidity / function. The principal burden is **chronic neurodevelopmental disability** — intellectual disability, language impairment, hypotonia, and epilepsy — with lifelong impact on independence, education, and communication. Morbidity is highly variable, from near-normal functioning (mild F-box variants) to severe multisystem involvement (structural/regulatory lesions).

Disease course / complications. Complications include epilepsy, GI dysfunction (constipation), recurrent respiratory/ENT infections, and the Wilms tumor risk. Recovery is not expected (static developmental disorder), but supportive therapy improves functional outcomes.

Prognostic factors. The strongest prognostic indicator identified is **variant/domain type** — F-box-domain variants portend milder outcomes than WD40-domain variants and structural/regulatory lesions. Presence of structural brain anomalies, early refractory epilepsy, and multisystem CNV/translocation lesions predict more severe courses. No validated molecular prognostic biomarker exists.

Section 12 — Treatment

No targeted or disease-modifying therapy exists. Management is **symptomatic and supportive**, individualized to the phenotype.

Domain	Intervention	NCIT suggestion
Developmental	Early intervention — physical, occupational, speech therapy	NCIT:C15265 (Rehabilitation Therapy)
Epilepsy	Anti-seizure medication (absence seizures often drug-responsive; PMID: 41685408)	NCIT:C264 (Anticonvulsant Agent)
Gastrointestinal	Management of constipation (diet, laxatives)	NCIT:C258 (Laxative)
Infections	Treatment/prevention of recurrent respiratory/ENT infections	NCIT antimicrobial classes as indicated
Oncologic	Wilms tumor surveillance; standard nephroblastoma therapy if tumor arises	NCIT:C17024 (Screening); NCIT:C15431 (Antineoplastic therapy)
Behavioral	Behavioral/psychological support	NCIT:C15277 (Behavioral Therapy)
Counseling	Genetic counseling	NCIT:C15417 (Genetic Counseling)

Pharmacogenomics / advanced therapeutics / surgery. No pharmacogenomic guidance, gene therapy, cell therapy, RNA therapy, or disease-specific surgical intervention is established. Surgery is relevant only for Wilms tumor (nephrectomy) or complication management.

Experimental directions. Because the disorder is driven by substrate accumulation in the NOTCH/MYC/CYCLIN E axis, pathway-targeted strategies (e.g., NOTCH inhibition) are mechanistically plausible but **entirely preclinical/unproven** for this germline disorder. In cancer contexts, FBXW7 loss is **synthetic lethal with CDC7 inhibition** (PMID: 37866880) — relevant to tumor therapeutics, not to the neurodevelopmental phenotype.

Section 13 — Prevention

Primary prevention. Not possible for de novo cases. For families with a known variant, reproductive options include **prenatal diagnosis** and **preimplantation genetic testing (PGT-M)**.

Secondary prevention — tumor surveillance. This is the most actionable prevention measure. Given the two-hit Wilms tumor risk, **renal/abdominal ultrasound surveillance** is recommended for individuals with constitutional/germline *FBXW7* pathogenic variants: *"Wilms tumor screening should be considered in individuals with constitutional or germline pathogenic variants in FBXW7 and associated neurodevelopmental syndrome"* (PMID: 38169111). Surveillance protocols typically follow established cancer-predisposition guidelines (e.g., periodic ultrasound through early childhood).

Tertiary prevention. Proactive management of epilepsy, aspiration/infection risk, constipation, and developmental support to prevent secondary complications and optimize function.

Counseling and screening. Genetic counseling addresses autosomal dominant inheritance, incomplete penetrance, variable (domain-dependent) expressivity, de novo vs inherited/mosaic origin, recurrence risk, and reproductive options. **Cascade genetic testing** identifies at-risk relatives who may warrant tumor surveillance.

Immunization / public health / environmental. No disease-specific immunization or public-health/environmental prevention applies (Mendelian disorder).

Section 14 — Other Species / Natural Disease

Taxonomy and orthologs. FBXW7 is deeply evolutionarily conserved. Orthologs: - **Mouse** (*Mus musculus*, NCBI Taxon 10090): *Fbxw7* / *Fbw7* (NCBI Gene 50754) - **Zebrafish** (*Danio rerio*, NCBI Taxon 7955): *fbxw7* - **Fruit fly** (*Drosophila melanogaster*, NCBI Taxon 7227): *archipelago* (*ago*) - **Nematode** (*Caenorhabditis elegans*, NCBI Taxon 6239): *sel-10*

Natural disease / veterinary relevance. No naturally occurring FBXW7-NDD-equivalent disorder in companion animals or wildlife is documented in OMIA within this investigation. *FBXW7* is studied experimentally across species but is not a recognized spontaneous veterinary disease.

Comparative biology / evolutionary conservation. The disease mechanism — loss of FBXW7 function → NOTCH/substrate accumulation → neural precursor dysregulation — is **conserved across vertebrates and invertebrates**, as demonstrated by the mouse cerebellar, zebrafish spinal-cord, and *Drosophila* learning phenotypes (Section 6; PMID: 21827743, PMID: 22554084, PMID: 35395208). *"fbxw7 … encodes the substrate recognition component of a ubiquitin ligase that targets Notch and other proteins for degradation"* (PMID: 22554084).

Transmission. Not applicable (non-infectious, non-zoonotic).

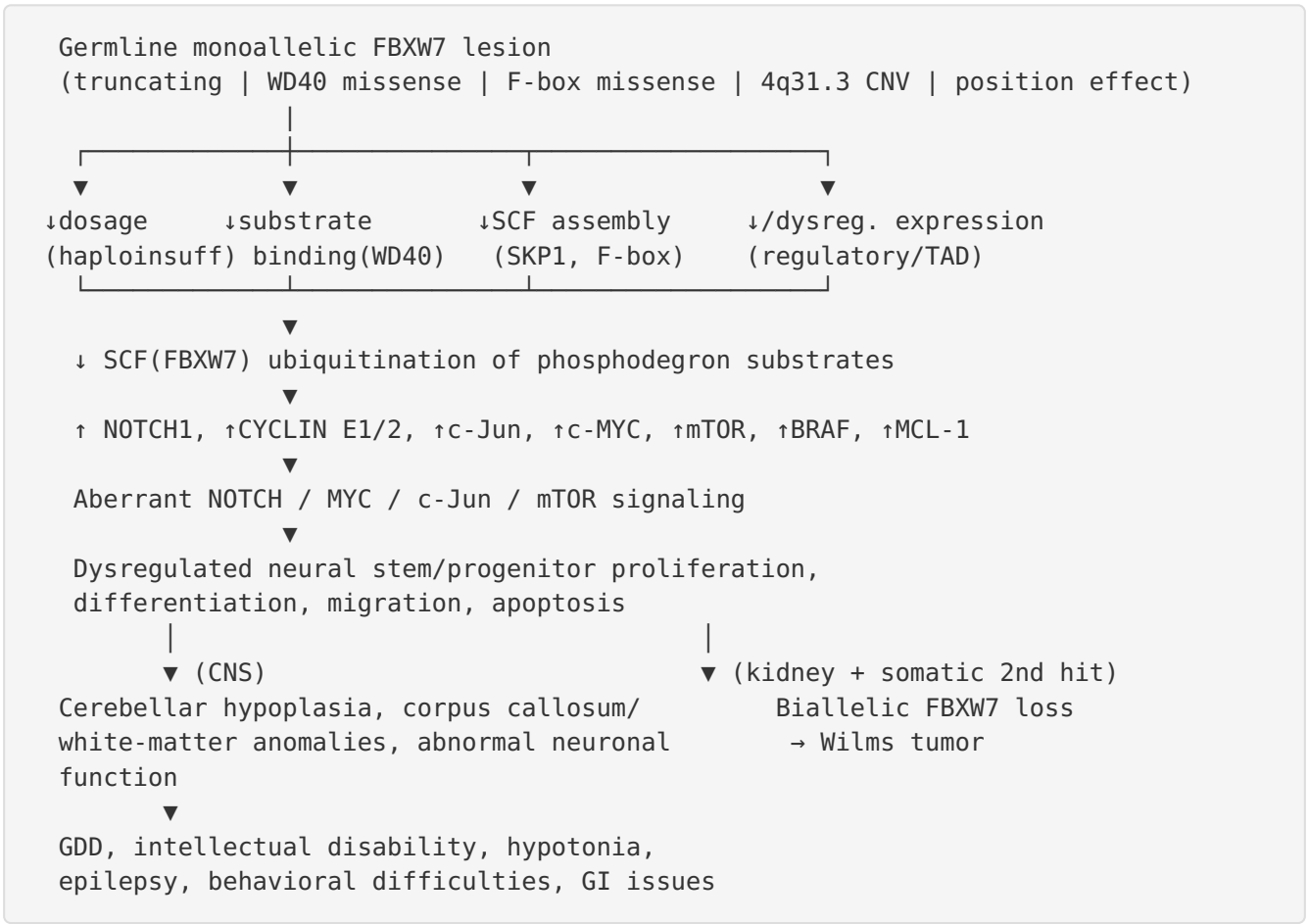
Section 15 — Model Organisms

Model	System / allele	Key phenotype	Human relevance	Reference
Mouse (cerebellar conditional KO, *Fbw7^ΔCb*)	Mammalian, conditional knockout	↓Cerebellar size, ↓Purkinje cells, defective axonal arborisation, supernumerary fissures, aberrant progenitor migration; ↑Notch1 & p-c-Jun; rescued by c-Jun deletion	Recapitulates human cerebellar anomalies; confirms substrate-accumulation mechanism	PMID: 21827743
Mouse (neural-lineage KO)	Mammalian, conditional knockout	Perinatal lethal; controls NSC differentiation/ apoptosis via Notch & c-Jun	Establishes essential developmental role	PMID: 21827743
Zebrafish (*fbxw7* mutant, vu56)	Vertebrate, germline mutant	Elevated Notch in spinal-cord precursors; excess oligodendrocyte progenitors; suppressible by Notch inhibition	Demonstrates Notch-dependent precursor-fate defect	PMID: 22554084
Drosophila (*archipelago* knockdown)	Invertebrate, pan-neuronal RNAi	Impaired learning and neuronal function	Links ortholog loss to behavioral/ cognitive deficit	PMID: 35395208
Cultured cells (recombinant variant expression)	In vitro	Impaired CYCLIN E1/E2 turnover with patient missense variants	Functional validation of pathogenicity	PMID: 35395208
MYCN gain-of-function mouse	Mammalian	Megalencephaly/ polydactyly via reduced FBXW7-mediated MYCN degradation	Illustrates FBXW7–substrate axis in brain growth (reciprocal direction)	PMID: 37710961

Genetic model types available. Knockout (constitutive embryonic-lethal), **conditional/tissue-specific knockout** (cerebellar, neural-lineage), germline mutant (zebrafish), and RNAi knockdown (Drosophila). Humanized/knock-in patient-variant models are not yet established but are a logical next step.

Recapitulation and limitations. Models faithfully reproduce the **cerebellar/neural-precursor and NOTCH/c-Jun-accumulation** core mechanism. Limitations: constitutive null is embryonic/perinatal lethal, requiring conditional approaches; models do not fully capture the human cognitive/behavioral spectrum, the domain-dependent expressivity, or the two-hit Wilms-tumor predisposition in a germline-heterozygous context. iPSC-derived neural organoids carrying patient variants would address human-specific developmental questions.

Mechanistic Model / Interpretation



The unifying theme is that **FBXW7 is a dosage-sensitive negative regulator of growth-promoting signaling in neural progenitors**. Any lesion that lowers functional FBXW7 — whether by reducing protein amount (truncating/deletion/position effect), crippling substrate capture (WD40 missense), or preventing ligase assembly (F-box missense) — converges on the same downstream outcome of **substrate accumulation and signaling excess**. The degree of residual function plausibly explains the phenotypic gradient, with F-box assembly-defect variants retaining more activity (milder phenotype) than WD40 substrate-binding variants. The same loss-of-function biology, when completed by a somatic second hit in renal tissue, explains the Wilms tumor predisposition — making FBXW7-NDD a rare example of a disorder that is simultaneously a neurodevelopmental syndrome and a cancer-predisposition syndrome.

Evidence Base

PMID	Study type	Contribution
35395208	Human cohort (n=35) + in vitro + Drosophila	**Foundational paper.** Defines core phenotype, variant spectrum, WD40 structural modeling, CYCLIN E1/E2 functional assay, Drosophila learning deficit
42432493	Human case series (n=7, 2026)	Phenotype frequencies (hypotonia 100%, GDD/language 86%, ID 57%, epilepsy 57%); extracerebral features; expanding variability
42111496	Review	Clinical spectrum, molecular mechanisms, and tumor predisposition overview
21827743	Mouse conditional KO	Cerebellar phenotype; Notch1/c-Jun accumulation; c-Jun rescue; NSC role
22554084	Zebrafish mutant	Notch-dependent neural precursor fate defect; conserved FBXW7 function
35515121	Review	FBXW7 substrate network (c-MYC, mTOR, MCL-1, Notch, cyclin E) in ubiquitin-proteasome system
35346215	Review	Phosphodegron-dependent ubiquitination mechanism; LoF in cancer
38169111	Human case report	Germline FBXW7 → NDD + Wilms tumor; surveillance recommendation
39414990	Human case report	Somatic second-hit in tumor DNA; two-hit model for Wilms tumor
41685408	Human case report	F-box-domain variant → mild phenotype; SKP1/SCF-assembly mechanism
38780184	Human case report + Hi-C	4q31.3 regulatory CNVs; TAD/dosage mechanism
39123068	Human case report	Translocation position effect reduces FBXW7 expression (haploinsufficiency)
37710961	Human + mouse	MYCN GoF escapes FBXW7 degradation → megalencephaly; validates FBXW7–substrate brain axis
37866880	CRISPR screen (cancer)	FBXW7 loss synthetic-lethal with CDC7 inhibition (therapeutic concept)

PMID	Study type	Contribution
38114583	Human cohort (India)	FBXW7 among de novo ID genes on neurodevelopmental panels

Evidence source distribution: The clinical phenotype rests on **human cohort and case-series data**; the mechanism rests on **model-organism (mouse, zebrafish, Drosophila) and in vitro** data plus human functional/structural modeling; the tumor link rests on **human case reports with molecular tumor analysis**.

Limitations and Knowledge Gaps

1. **Small sample sizes.** The entire literature comprises <50 reported individuals. Phenotype frequencies (from a 7-person series) have wide confidence intervals and may not generalize.
2. **Penetrance and prevalence unquantified.** True population prevalence, incidence, penetrance, and the proportion of inherited vs de novo cases are not precisely known.
3. **Genotype–phenotype correlation is preliminary.** The F-box-vs-WD40 severity contrast rests largely on single cases; a systematic domain-stratified analysis is lacking.
4. **No validated biomarker or episignature.** Diagnosis is purely molecular; no biochemical marker, imaging signature, or methylation episignature aids diagnosis or prognosis.
5. **Tumor risk magnitude unknown.** The absolute lifetime Wilms tumor risk (and whether other tumors occur) is unquantified, so surveillance intensity/duration is not evidence-based.
6. **Missing ontology/registry identifiers.** OMIM phenotype number, Orphanet number, and ICD codes were not confirmed in this investigation and need direct verification.
7. **Mechanistic gaps in human brain.** Human-specific neurodevelopmental consequences are inferred from animal models; iPSC/organoid data are absent.
8. **No therapeutics evaluated.** No clinical trials; pathway-targeted strategies are purely hypothetical.

Proposed Follow-up Experiments / Actions

1. **Build a prospective patient registry** to quantify penetrance, phenotype frequencies, natural history, and absolute Wilms tumor risk — directly informing surveillance guidelines.
2. **Systematic domain-stratified genotype–phenotype analysis** (F-box vs WD40 vs truncating vs regulatory) across all reported variants, with standardized functional assays (substrate turnover, SCF assembly) to test the residual-function–severity hypothesis quantitatively.
3. **Patient-derived iPSC neural organoids** carrying representative variants to measure NOTCH/CYCLIN E/c-Jun accumulation and progenitor-fate defects in a human developmental context.
4. **Develop a DNA-methylation episignature** from patient blood to provide a diagnostic/variant-classification aid.
5. **Define evidence-based tumor surveillance** (ultrasound frequency, age range) via multicenter data pooling, modeled on existing cancer-predisposition protocols.
6. **Verify and register ontology identifiers** (OMIM, Orphanet, ICD-11, HPO annotations with frequencies) for knowledge-base completeness.
7. **Preclinical pathway-rescue studies** (e.g., NOTCH or c-Jun modulation) in conditional mouse and zebrafish models to assess whether developmental defects are reversible within critical windows.

Consensus Answer

FBXW7-Related Neurodevelopmental Disorder (MONDO:0859280) is a rare autosomal dominant Mendelian syndrome caused by monoallelic loss-of-function variants (truncating, WD40- or F-box-domain missense, deletions, or 4q31.3 regulatory lesions) in *FBXW7*, the substrate-recognition subunit of the SCF E3 ubiquitin ligase; impaired ubiquitination and degradation of substrates such as NOTCH1, CYCLIN E1/2, and c-Jun dysregulates neural progenitor proliferation and differentiation, producing global developmental delay, intellectual disability, hypotonia, epilepsy, variable cerebellar/corpus-callosum/white-matter anomalies, and gastrointestinal issues. The disorder shows incomplete penetrance and variable, partly domain-dependent expressivity (F-box variants milder than WD40), carries a

Wilms-tumor predisposition via a germline-plus-somatic two-hit mechanism warranting renal ultrasound surveillance, and is managed supportively with no targeted therapy currently available.
